Can Data Selection and Preprocessing Improve Image–Text Retrieval?

CSCI 5942: AI Engineering | Mega Assignment Proposal | Fall 2026

Andrew Wernersbach, Bill Hsu

1 What will we build?

We propose to investigate whether dataset selection and input preprocessing can improve image–text retrieval for a vision–language model (VLM) trained on image-caption pairs from the 2017 Microsoft Common Objects in Context (COCO) dataset [1]. The VLM will be randomly initialized and trained from scratch using a fixed budget of 15,000 training images per run.

We will study five dataset-selection methods and five input/preprocessing configurations in two stages. First, nine distinct training runs will isolate the effects of selection and preprocessing relative to a shared baseline. Based on their retrieval performance, we may then evaluate up to five additional, promising selection–preprocessing pairings. This approach prioritizes informative comparisons within the compute budget. The methods are described in Appendix A.

2 On what data?

Training will use COCO train2017, with exactly 15,000 unique images per run. Each COCO image has five captions [1]; one caption will be sampled per image each epoch so different descriptions can contribute without changing the image budget.

The initial nine runs consist of five selection methods evaluated with standard RGB preprocessing and five preprocessing methods evaluated with random image selection. The random-selection/standard-RGB baseline is shared between these groups and trained only once. Up to five follow-up runs will combine methods selected on the basis of initial retrieval performance. Each run uses one weight initialization and 40 training epochs.

Evaluation will use the full held-out COCO val2017 split of approximately 5,000 images, each with five reference captions. All runs will be evaluated on this same split.

3 On what compute?

We estimate compute using a ViT-B/16 image encoder at 640×640 resolution and a 12-layer Transformer text encoder. A forward pass costs approximately 371.03 GFLOPs per image–caption pair; training is approximated as three times this amount. Detailed arithmetic is given in Appendix B.

For 15,000 images and 40 epochs per run, the estimated training compute is

$$F_{\text{run}} \approx 15{,}000 \times 40 \times 3 \times 371.03 \times 10^9 \approx 6.68 \times 10^{17} \text{ FLOPs}, \tag{1}$$

$$F_{\text{total}} \approx (9+k)F_{\text{run}}, \qquad 0 \le k \le 5, \tag{2}$$

$$\approx \boxed{(6.01\text{–}9.35) \times 10^{18} \text{ FLOPs}}. \tag{3}$$

Here k denotes the number of optional follow-up runs. The estimate excludes one-time preprocessing, frozen CLIP feature generation, data loading, and other system overhead.

4 How will we know this worked?

We will measure image–text retrieval on the held-out COCO val2017 split. Both image-to-text and text-to-image retrieval will be evaluated with Recall@1, Recall@5, and Recall@10. For image-to-text retrieval, any of an image's five reference captions is a valid match; for text-to-image retrieval, the associated image is the correct match.

We will first compare each of the nine initial runs against the shared random-selection/standard-RGB baseline, holding the training-image and epoch budgets fixed. These results will identify which selection and preprocessing methods are most promising. If warranted, up to five additional runs will test selected combinations against their individual-method results and the same baseline, allowing us to investigate whether their benefits reinforce one another.

References

[1] T.-Y. Lin, M. Maire, S. J. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common Objects in Context," in *European Conference on Computer Vision (ECCV)*, 2014, pp. 740–755.

[2] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning Transferable Visual Models From Natural Language Supervision," in *Proceedings of the 38th International Conference on Machine Learning*, vol. 139, 2021, pp. 8748–8763.

[3] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," in *International Conference on Learning Representations (ICLR)*, 2021.

A Dataset Selection and Preprocessing Methods

The initial stage tests the five dataset-selection methods using standard RGB preprocessing and the five input/preprocessing configurations using random image selection. Because random selection with standard RGB is the shared baseline, this requires nine distinct training runs. After evaluating retrieval performance, up to five further runs may combine the most promising selection and preprocessing methods. Each run uses 15,000 unique training images and 40 epochs; the full study therefore contains 9–14 runs.

A.1 Dataset Selection Methods

1. **Random selection:** randomly select 15,000 images from COCO train2017. This provides the baseline against which the other selection strategies are compared.

2. **Object-category-balanced selection:** select images to improve coverage of the 80 COCO object categories while maintaining the 15,000-image budget.
3. **Visual-diversity selection:** use frozen pretrained CLIP image embeddings to select a visually diverse subset of images, reducing redundancy in the selected training data [2].
4. **Caption-semantic-diversity selection:** use text embeddings of COCO captions to select images whose captions provide diverse semantic coverage.
5. **Quality-aware diversity selection:** incorporate basic image-quality measures such as resolution, blur, and visual complexity while maintaining diversity in the selected subset.

These methods represent different hypotheses about what makes a small training subset useful: random sampling provides a baseline, category balancing emphasizes object coverage, visual diversity emphasizes visual coverage, caption diversity emphasizes linguistic coverage, and quality-aware selection emphasizes the quality and diversity of individual examples.

A.2 Input and Preprocessing Configurations

1. **Standard RGB:** resize the RGB image directly to 640×640.
2. **Aspect-ratio-preserving RGB:** resize the image while preserving its original aspect ratio and pad the resulting image to 640×640.
3. **Random resized crop:** apply random resized cropping during training before producing a 640×640 input.
4. **Color and contrast normalization:** apply standardized color and contrast normalization before producing the RGB input.
5. **RGB + CLIP features:** provide the processed RGB image together with a frozen 512-dimensional CLIP image embedding as an auxiliary feature [2].

The CLIP features in the fifth configuration are generated by a pretrained model and remain frozen during VLM training. Thus, the primary VLM is still trained from randomly initialized weights; the experiment instead measures whether pretrained visual information is useful when provided as an auxiliary input.

B Compute Arithmetic

The compute estimate assumes a ViT-B/16 image encoder operating at 640×640 resolution and a 12-layer Transformer text encoder.

At 640×640 resolution, ViT-B/16 processes

$$N_{\text{patches}} = \frac{640}{16} \times \frac{640}{16} \tag{4}$$

$$= 40 \times 40 \tag{5}$$

$$= 1600 \tag{6}$$

image patches. The image encoder forward-pass cost is approximated as

$$F_{\text{image}} \approx (1.89 + 366.16) \times 10^9 \tag{7}$$
$$\approx 368.05 \times 10^9 \text{ FLOPs/image.} \tag{8}$$

For the text encoder, assuming hidden dimension $d = 512$, intermediate dimension $d_{\text{ff}} = 2048$, sequence length $L = 77$, and 12 Transformer layers,

$$F_{\text{text,layer}} \approx 4Ld^2 + 2L^2d + 2Ldd_{\text{ff}}. \tag{9}$$

This gives approximately

$$F_{\text{text}} \approx 2.98 \times 10^9 \text{ FLOPs/text.} \tag{10}$$

Therefore, the combined forward-pass cost for one image-caption pair is

$$F_{\text{forward}} = F_{\text{image}} + F_{\text{text}} \tag{11}$$
$$\approx 371.03 \times 10^9 \text{ FLOPs.} \tag{12}$$

We approximate training computation as three times the forward-pass computation:

$$F_{\text{training}} \approx 3F_{\text{forward}}. \tag{13}$$

The initial design contains five selection-only and five preprocessing-only configurations, with their shared baseline counted once. Let k be the number of follow-up runs, with $0 \leq k \leq 5$. Every run uses one initialization, 15,000 images, and 40 epochs. Thus,

$$N_{\text{initial}} = 5 + 5 - 1 = 9, \tag{14}$$
$$N_{\text{runs}} = 9 + k, \qquad 0 \leq k \leq 5, \tag{15}$$
$$F_{\text{run}} \approx 15{,}000 \times 40 \times 3 \times 371.03 \times 10^9 \tag{16}$$
$$\approx 6.67854 \times 10^{17} \text{ FLOPs,} \tag{17}$$
$$F_{\text{initial}} \approx 9F_{\text{run}} \approx 6.01 \times 10^{18} \text{ FLOPs,} \tag{18}$$
$$F_{\text{maximum}} \approx 14F_{\text{run}} \approx 9.35 \times 10^{18} \text{ FLOPs.} \tag{19}$$

These are theoretical training-compute estimates, excluding data loading, image preprocessing, optimizer overhead, hardware inefficiency, and the one-time cost of generating frozen CLIP embeddings.